	Since average kinetic energy increases, the temperature increases.
12	**D** Positive work is done on the gas by its surrounding during compression, when the pressure is 1.0×10^5 Pa. $WD_1 = -p(V_f - V_i) = -1.0 \times 10^5 \times (2 - 5) \times 10^{-4} = 30$ J Negative work is done on the gas by its surrounding during expansion, when the pressure is 1.5×10^5 Pa. $WD_2 = -p(V_f - V_i) = -1.5 \times 10^5 \times (5 - 2) \times 10^{-4} = -45$ J Net work done on gas = 30 – 45 = – 15 J.
13	**D**